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A Project Report

On

# Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations

Submitted in partial fulfilment of the requirement of

**University of Mumbai**

For the Degree of

**Bachelor of Technology (B.Tech.)**

*in*

**Computer Engineering**


*Submitted by*

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**Department of Computer Engineering**

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**An Autonomous Institute Permanently Affiliated to University of Mumbai**

**Sector 9A, Vashi, Navi Mumbai – 400703**


**UNIVERSITY OF MUMBAI**

**2026-27**

# APPROVAL SHEET

This is to certify that the project entitled

**"Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations"**

**Submitted by**

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**2.** _____

**Head of Department:** _____

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**Place :**

# CERTIFICATE

This is to certify that the project entitled

**Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations**

**Submitted by**


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In partial fulfilment of the requirements for the degree of **B.Tech. in Computer Engineering**, the Semester VII Major Project is approved.

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**College Seal**

**Date:** _____

# Declaration

We declare that this written submission for BE Major Project Declaration entitled "**Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations**" represent our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhere to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by institute and also evoke penal action from the sources which have thus not been properly cited or from whom paper permission have not been taken when needed.

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# Abstract

Business teams in HR, marketing and podcast production spend much of their time on repetitive work such as screening resumes, rewriting posts for each channel and researching guests. Existing tools mostly store records or match keywords, and their AI features give opaque results that are difficult to explain or audit. Our project aims to build an integrated agentic AI platform in which a workflow router coordinates specialised LLM agents for recruitment, marketing, podcast guest research and enterprise integrations. It combines retrieval-augmented generation with rubric-based, evidence-backed scoring and secure connectors to Zoho, Workday and Greenhouse-style systems. The system generates rubrics from job descriptions, screens and interviews candidates, drafts channel-specific content and prepares source-backed guest briefings. By placing human approvals, PII protection and audit logs around consequential actions, the platform helps organisations automate routine work while keeping people in control of final decisions.

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# Chapter 1

# Introduction

## 1.1 Background

Most business teams today still spend a large part of their day on work that is repetitive but hard to fully delegate. An HR recruiter might sift through hundreds of resumes for a single role, spending hours on a task that follows the same mental checklist every time. A marketing team rewrites the same campaign message five times, once for LinkedIn, once for Instagram, once for a newsletter, and so on. A podcast producer spends an afternoon building a guest briefing by piecing together interviews, articles and social profiles from across the web. None of this requires deep creativity, yet it consumes the time people need for the decisions that actually matter.

The software these teams use was not built to help much. Applicant tracking systems, HR platforms and CRM tools are good at storing and retrieving records, but they struggle with anything unstructured, such as a resume, a job description written in natural language, or a long-form article. Their rules are fixed, which means they cannot adapt when the inputs do not match the expected format. Recent advances in large language models have changed what is possible here [1, 2, 3]. These models can read unstructured text, reason about it, call external tools and pass results to other agents in a pipeline. Our project, developed in partnership with InfinityPool (Shankh), applies this capability to the everyday business workflows where the gap between what software currently does and what teams actually need is widest.

## 1.2 Motivation

The problems with today's tools are not just inconveniences. In hiring, keyword-based screening has been shown to reward candidates who know how to phrase a resume rather than those who are actually the best fit for the role [4]. Worse, AI screening tools carry measurable racial and gender

bias [5], and some can be fooled by hidden text that a human reader would never notice [6]. When a recruiter receives a ranked shortlist with no explanation, they have no way to catch these errors before they affect a real person's career.

In marketing, content that goes out without a proper review can damage a brand in ways that take months to repair. In podcast research, a briefing built from unsourced or misattributed claims creates serious reputational risk for the host and the show [10, 8]. And across all three functions, the data needed to do the work sits in separate systems, each with its own login, its own API, its own rate limits and its own quirks. Getting everything into one place is itself a full-time job.

What teams really need is a system that can handle the repetitive parts end to end, explain every output clearly enough that a person can review and trust it, and then wait for that person to give the final go-ahead before anything consequential happens. That is the gap this project is designed to close.

## 1.3 Aim and Objectives

**Aim :**

The aim of **Business Function Agentic Workflows** is to design and build an integrated multi-agent AI platform that takes over the repetitive, high-volume parts of HR recruitment, marketing content operations, podcast guest research and enterprise data integration, while keeping a human reviewer in control of every decision that matters. The platform uses specialised LLM agents coordinated by a workflow router, grounded by a retrieval-augmented knowledge layer, and governed by approval gates and audit logs throughout.

**Objectives :**

The specific objectives of the project are:

- To build HR workflows that generate weighted rubrics from a job description, screen and score resumes with cited evidence, conduct AI-assisted interviews and produce a ranked shortlist for the recruiter to review, alongside assistants that answer onboarding, leave and HR policy questions.
- To build marketing workflows that turn a theme or source content into a content calendar, channel-specific drafts and audience segments,

with every piece of content held for human approval before it goes anywhere.

- To build podcast research workflows that gather public sources about a guest, index them for retrieval, and produce a structured briefing in which every claim is linked to a source and labelled as a verified fact, an allegation or an opinion.
- To build a connector layer that communicates with Zoho, Workday-style and Greenhouse-style systems through authenticated, rate-limit-aware, idempotent API calls, with human approval required before any write operation goes through.
- To enforce consistent governance across all workflows through a compliance reviewer that checks for PII, prompt injection and unsupported claims, an approval queue that holds every consequential output, and an audit log that records every prompt, tool call, source and reviewer action.
- To package the entire platform as a reusable workflow framework and measure its performance on accuracy, hallucination rate, latency, cost and task completion.

# Chapter 2

# Study Of the System

## 2.1 Literature Survey

Building an agentic AI platform for business functions means drawing on work from several overlapping areas of research. We looked at how multi-agent frameworks are structured, how AI has been applied to hiring and whether it can be trusted to do so fairly, how marketing content generation and hallucination reduction have been approached, and what the state of the art says about tool use, privacy and governance. The picture that emerges is one of genuine progress alongside well-documented gaps, and those gaps are what this project tries to close.

**Multi-Agent Orchestration.** Wu et al. [1] introduced AutoGen, a framework that lets developers compose applications from multiple agents, each backed by an LLM, a human or an external tool. Guo et al. [2] followed with a broad survey of LLM-based multi-agent systems, covering how agents are profiled, how they communicate and how their capabilities develop over time. Yao et al. [3] showed that alternating between reasoning steps and tool actions produces better results than either alone. What none of these works provide is a ready-made path to a governed, domain-specific business workflow. They give us building blocks, but the plumbing for rubric-based scoring, human approval gates and decision-level audit logs still has to be designed from scratch.

**AI-Powered Recruitment and Bias.** The recruitment literature makes a sobering case for caution. Gan et al. [4] showed that an LLM-based pipeline can screen resumes eleven times faster than a human and reach a respectable F1 score, but their system was tested only on IT-sector data and could not generate rubrics from a job description or conduct an interview. Wilson and Caliskan [5] found that embedding-based screening favoured White-associated names in 85 out of every 100 cases and that Black male candidates were disadvantaged in up to 100% of test cases. Mu

et al. [6] then showed that adversarial content hidden inside a resume could manipulate LLM screening more than 80% of the time, and that the best available defences reduced this by only around a quarter. Stuart et al. [7] took things further by building an AI interviewer that updates its belief about a candidate's competencies after each answer, recovering accurate ability profiles with 76.1% accuracy. Together these papers describe a picture where AI recruitment tools can save time but cannot yet be trusted without human oversight, bias monitoring and adversarial scanning at every step.

**Content Generation and Marketing.** Liu et al. [8] demonstrated that retrieval-augmented generation can produce keyword-specific ad copy that outperforms template-based content in online A/B tests, with an automated evaluation approach that agrees with human reviewers nearly 90% of the time. The limitation is scope: the system was designed for search advertising rather than for brand-consistent posts across multiple social channels, and it does not include an approval step before content goes live. Our marketing workflow extends the RAG-based generation idea to channel-specific branded content and adds an explicit queue where a human must approve every draft before it is published.

**Retrieval-Augmented Generation and Hallucination Reduction.** Asai et al. [9] trained a model to decide for itself when to retrieve and to critique its own outputs using special reflection tokens, producing better factuality scores than baseline retrieval-augmented approaches. Dhuliawala et al. [10] took a different route, using the model to draft an answer, generate verification questions and then revise the answer based on the results, reducing hallucinations on several benchmarks. Both approaches have a real limitation in practice: they either need custom training or rely solely on what the model already knows, without the ability to consult an external source for a recent or obscure fact about a real person. Our podcast research agent borrows the retrieve-then-verify idea and applies it through prompting, requiring a public source for every claim and labelling each one as a verified fact, an allegation or an opinion before anything reaches the host.

**Tool Use and Enterprise API Integration.** Schick et al. [11] showed that a language model can learn which API to call and when, purely from a small number of demonstrations, without supervised training. Patil et al. [12] went further, fine-tuning a model to write accurate API calls even when the documentation changes after training. Both results are impressive, but

neither system can chain multiple authenticated calls, handle rate limits or decide whether a particular write operation should require human approval before it goes through. Enterprise integrations with Zoho, Workday and Greenhouse require all of these things, which is why our connector layer adds OAuth scopes, retries, idempotency keys and a mandatory approval step for any call that writes data.

**Privacy and Governance.** Lukas et al. [13] measured how much personally identifiable information can leak from a language model and found that even techniques like differential privacy leave roughly 3% of PII sequences exposed. Mökander et al. [14] proposed a governance blueprint for LLMs that operates at the organisational, model and application levels, but left the question of how to implement application-level auditing in software to the reader. Our system addresses the run-time PII risk directly by redacting candidate data before any LLM call and restricting what appears in logs. The application audit layer is not a design concept here but a concrete implementation: every prompt, tool call, source, draft version, reviewer decision and final output is stored and traceable.

## 2.2 Existing Systems

A number of commercial tools already address parts of the problem this project is trying to solve. Looking at them side by side makes the gap clearer.

**HireVue AI Interviewer** runs two-way structured interviews and scores responses against role-specific criteria with explanations. It does one thing well but nothing else. It does not generate rubrics from a job description, it does not check resume claims for accuracy, and marketing or research workflows are completely outside its scope. Because it is proprietary, its design cannot be studied or extended.

**Greenhouse AI Recruiting** is built around structured hiring processes. Its AI features handle resume parsing with anonymisation, generate interview stages and produce scorecard summaries. The AI is there to support a human interviewer, not to run a pipeline end to end. There is no evidence verification against external sources, and nothing in the platform touches marketing or research.

**Zoho Recruit with Zia** does a good job of matching candidates to jobs, drafting job descriptions and emails, and summarising recorded interview transcripts. What it cannot do is conduct a live adaptive AI interview, produce rubric-based scores with cited evidence and an approval gate, or handle any workflow outside recruiting.

**Jasper** is a capable marketing workspace that stores brand voice, style guides and audience profiles and uses them to guide content generation. Its coverage is marketing only. Its governance and approval workflows are not publicly documented, and it does not produce source-cited research briefings of any kind.

**n8n** is a flexible workflow automation platform with a visual builder and a large library of integrations including AI-agent nodes. It is an excellent toolkit but it is only a toolkit. Domain-specific logic for HR scoring, brand rules, research guardrails, PII redaction and claim labelling all have to be built separately by whoever sets it up.

**Perplexity Deep Research** is a fast and thorough general-purpose research tool. It does not format its output for a podcast host, it does not label claims as facts, allegations or opinions, and it provides no approval step or audit trail before a report is used.

The research literature summary table follows on the next page.

| S.no | Paper | Main Idea / Objective | Tools / Tech Used | Gaps Identified |
|---|---|---|---|---|
| 1 | AutoGen (2024) | Framework for building applications from multiple conversable agents that combine LLMs, humans and tools. | Multi-agent conversation framework; human-in-the-loop support. | General-purpose; no HR or marketing workflow, rubric scoring, approval gates or audit design. |
| 2 | LLM Based Multi-Agents: A Survey (2024) | Reviews domains, agent profiling, communication, capability growth and benchmarks of LLM multi-agent systems. | Systematic literature review; open-source tracking repository. | Survey only; no concrete business-function system, governance or integration. |
| 3 | ReAct (2023) | Interleaves reasoning traces with tool actions to reduce hallucination and improve decision-making. | Few-shot prompting; Wikipedia API; HotpotQA, FEVER, ALFWorld, WebShop. | Single agent with read-only tools; no approval or audit for actions with real effects. |
| 4 | LLM Agents in Recruitment (2024) | Agent pipeline that classifies, grades, summarises and selects resumes. | Fine-tuned LLaMA2; GPT-3.5 and GPT-4 baselines; BERT, T5. | IT resumes only; labels mostly from GPT-4; no JD-to-rubric generation or interview stage. |
| 5 | Bias in Resume Screening via LM Retrieval (2024) | Audits embedding-based screening for gender, race and intersectional bias. | Massive Text Embedding models; retrieval-based audit on 500+ resumes. | Measures bias; proposes no bias-aware screening or human review step. |
| 6 | Adversarial Vulnerabilities in Resume Screening (2025) | Shows hidden instructions and invisible content can manipulate LLM screening. | 12 LLM variants; prompt defence; FIDS (LoRA). | Defences cut attacks by only 10–26%; no pipeline-level scanning or human checkpoint. |
| 7 | Beyond the Resumé (2026) | Rubric-aware LLM interviewer that updates beliefs about competencies after each answer. | GPT-5; belief updating; metamorphic testing; Chainlit. | Simulated interviews only; three-level rubrics; no rubric generation, ATS link or HR approval. |
| 8 | LLMs for Marketing Content Generation (2025) | Generates keyword-specific ad copy and evaluates it at scale. | RAG; LLM-as-a-Judge; rule-based metrics; A/B testing. | Ad copy only; no multi-channel brand posts or approval workflow. |
| 9 | Self-RAG (2024) | Model learns when to retrieve and critiques its own output with reflection tokens. | Fine-tuned Llama2 7B/13B; retriever; reflection tokens. | Needs custom training; not evaluated on high-risk research about real people. |

| S.no | Paper | Main Idea / Objective | Tools / Tech Used | Gaps Identified |
|---|---|---|---|---|
| 10 | Chain-of-Verification (2024) | Drafts, plans and answers verification questions, then revises to reduce hallucination. | Multi-stage prompting; factored verification. | Uses only the model's own knowledge; no external sources or claim labels. |
| 11 | Toolformer (2023) | LLM teaches itself when and how to call external APIs. | GPT-J 6.7B; self-supervised API-call annotation. | One call per example; no tool chaining, authentication or write actions. |
| 12 | Gorilla (2024) | Fine-tuned LLM writes accurate API calls with retrieval over documentation. | LLaMA-7B; BM25 and GPT-Index retrievers; APIBench; AST matching. | Public ML APIs only; ignores permissions, PII and approval of write operations. |
| 13 | PII Leakage in Language Models (2023) | Defines and measures PII extraction, reconstruction and inference attacks. | GPT-2 fine-tuning; black-box attacks; differential privacy. | Covers training-data leakage; not PII in run-time prompts, logs and tools. |
| 14 | Auditing LLMs: A Three-Layered Approach (2024) | Blueprint of governance, model and application audits for LLMs. | Conceptual governance framework. | Conceptual; no implementation of application-level audit trails. |

Summary of Related Research Papers

# Chapter 3

# Proposed System

## 3.1 Problem Statement

The tools that business teams use today were built for a world where information lived in structured databases and workflows followed fixed, predictable rules. Hiring meant posting a job, collecting applications in a spreadsheet and reading through them one by one. Marketing meant creating a template and filling it in for each channel. Research meant opening several browser tabs and taking notes. These processes have not fundamentally changed, even as the volume of work has grown substantially and the stakes of getting things wrong have become clearer.

The consequences are real. A recruiter who screens a thousand resumes with a keyword filter will miss strong candidates who did not phrase their experience the right way. A screening system that assigns scores without explaining them gives the recruiter no way to catch mistakes or bias. A marketing team that publishes content without a structured review process is one bad post away from a brand crisis. A podcast host who builds a guest briefing from unchecked sources risks amplifying false or defamatory information on air. And in every case, the data the team needs is locked in a different system from the one they are currently looking at, each requiring its own login and its own way of being queried.

What is missing is a system that takes over the repetitive, high-volume parts of these workflows, explains every output clearly enough that a person can check it, and then waits for that person's approval before doing anything that cannot be undone.

## 3.2 Scope

This project covers four business workstreams, defined in collaboration with our industry partner InfinityPool (Shankh): HR recruitment and HR assistance, marketing content operations, podcast guest background re-

search, and enterprise integrations with Zoho, Workday and Greenhouse-style platforms. Across all four, the project includes a shared workflow router, a retrieval-augmented knowledge layer, human approval gates at every consequential step, PII protection and a complete audit log, all packaged as a reusable framework.

There are clear boundaries around what the system does and does not do. It assists people; it does not make final hiring, rejection, salary or disciplinary decisions on their behalf. It holds content for approval; it does not publish anything without a person saying yes. All development is done with synthetic, public or sanitised data and with mock or sandbox APIs, not live production credentials. The framework is designed to be extended to other business functions later, such as finance and sales, but that is outside the current scope.

## 3.3 Proposed System

**Business Function Agentic Workflows** is a modular multi-agent platform built around a single idea: business workflows should be fast and consistent where the work is repetitive, and human where the decisions matter.

### System Overview

Every interaction starts at a web dashboard. A user or an incoming business event triggers a request, which a workflow router classifies and sends to the right group of specialist agents. Those agents pull context from a shared knowledge base and call enterprise connectors as needed. Anything that involves a consequential action – sending an email to a candidate, publishing a post, writing a record back to an HR system – goes into an approval queue rather than executing immediately. A human reviewer looks at it, approves or edits it, and only then does the system act. The whole exchange is written to an audit log.

### Key Components

- **Web Dashboard** – A Next.js interface where HR managers, marketing teams and researchers interact with the system: creating jobs, reviewing candidate rankings, editing content drafts and clearing the approval queue. Candidates have a separate public-facing apply page.
- **Workflow Router** – Classifies each incoming request as an HR, marketing, podcast research or integration task and hands it to the match-

ing set of agents.

- **HR Agents** – A set of specialised agents covering rubric generation from job descriptions, resume parsing and adversarial scanning, claim verification, evidence-backed screening, interview question generation, live AI interviewing, candidate ranking and follow-up. Separate assistants handle employee queries about onboarding, leave, attendance and HR policy.
- **Marketing Agents** – Agents that turn a theme or source content into a content calendar, campaign and creative briefs, audience segments and channel-specific social copy. All drafts wait in an approval queue before going anywhere.
- **Podcast Research Agents** – Agents that gather public sources about a guest, index them for retrieval, build a structured briefing and label every claim as a verified fact, an allegation or an opinion. Sensitive-topic guardrails are applied before the briefing reaches the host.
- **Integration Agent and Connector Layer** – A unified connector interface for Zoho People, Zoho CRM and Recruit, Workday-style HRIS, Greenhouse-style ATS, CRM and email, calendar and social media APIs. Handles OAuth, rate limits, pagination, webhooks, retries and idempotency so that agents do not need to know the details of each system.
- **Knowledge (RAG) Layer** – A vector store holding company policies, brand guidelines, job descriptions, approved rubrics, previous campaigns and guest research sources. Agents retrieve from this store rather than working from memory alone.
- **AI Provider Layer** – A switchable configuration that routes agent calls to Google Gemini, OpenAI GPT models or local models through Ollama, depending on availability and cost.
- **Compliance Reviewer and Guardrails** – Sits between every agent output and the approval queue. Checks for PII that should not be in a prompt or a log, for prompt injection in user-uploaded content and for claims that are not supported by a cited source. Flags issues for human review rather than letting them through silently.

- **Approval Layer** – Holds rubrics, candidate decisions, content drafts, guest briefings and outgoing connector writes until a human acts on them. Nothing consequential executes without this step.
- **Audit Layer and Data Store** – Supabase-backed storage for all jobs, candidates, scores, drafts and reports, plus a structured log of every prompt, tool call, source, draft version, reviewer identity and final output. Protected by role-based access controls.

### Workflow of the Proposed System

When a request arrives, the workflow router classifies it and dispatches it to the appropriate specialist agents. The agents retrieve what they need from the knowledge base and call enterprise connectors where necessary. Every output that carries a consequence is held in the approval queue rather than acted on immediately. The compliance reviewer checks each item for PII exposure, prompt injection and unsupported claims before it reaches the reviewer. The human reviewer reads it, approves, edits or rejects it. Approved actions are sent through the connector layer. Every step from the initial prompt to the final output is written to the audit log.

# Chapter 4

# Design Of the System

## 4.1 System architecture

### 4.1.1 Data Flow Diagrams

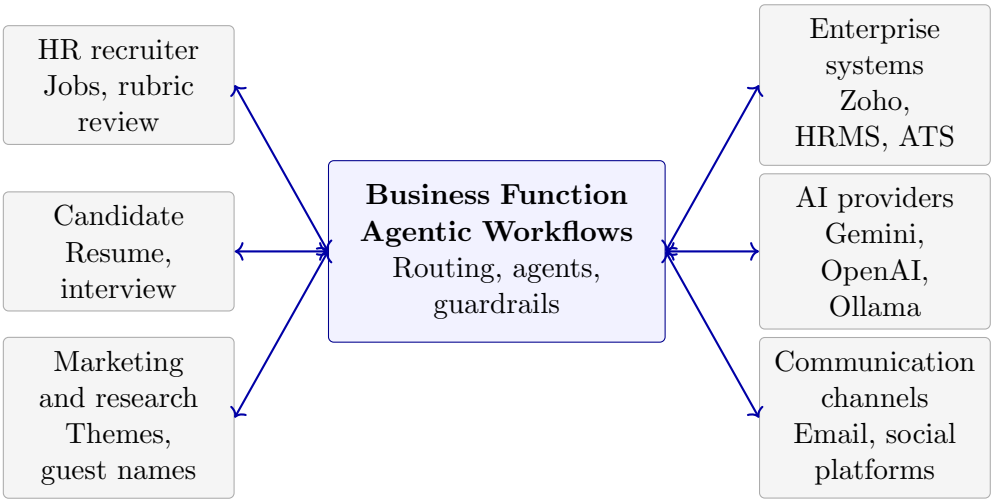



**Figure 4.1.1 DFD Level 0 of the System**

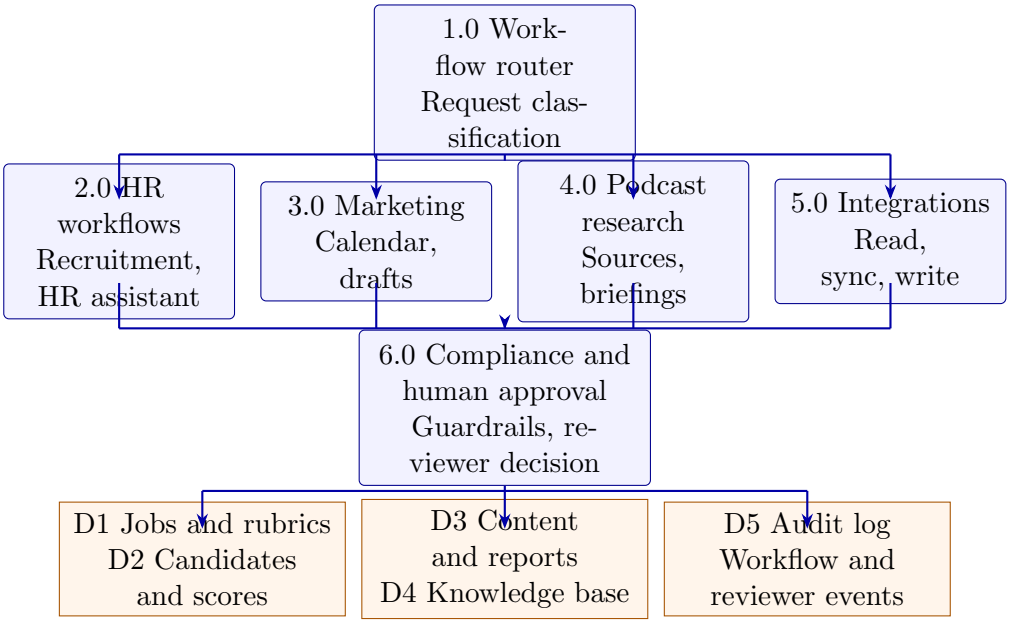



**Figure 4.1.2 DFD Level 1 of the System**

### 4.1.2 UI/UX Diagram

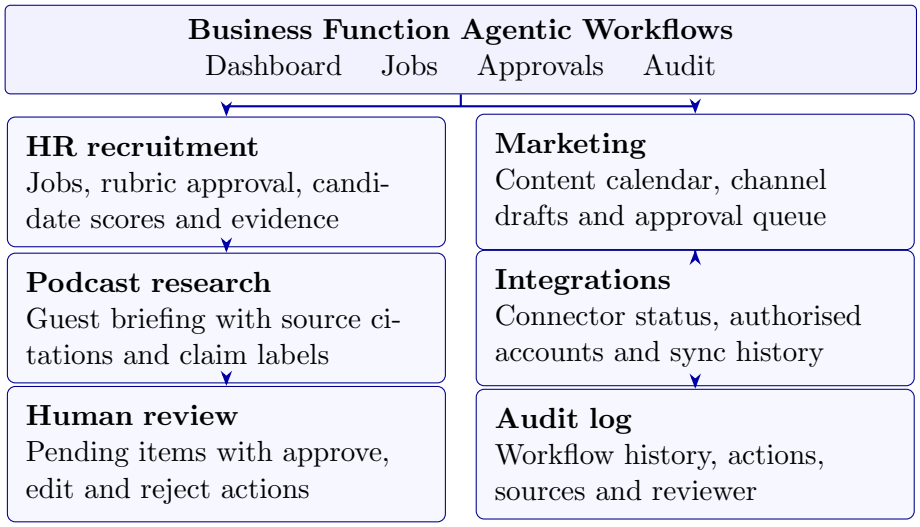



**Figure 4.2.1 UI/UX Diagram of the System**

### 4.1.3 Block Diagram

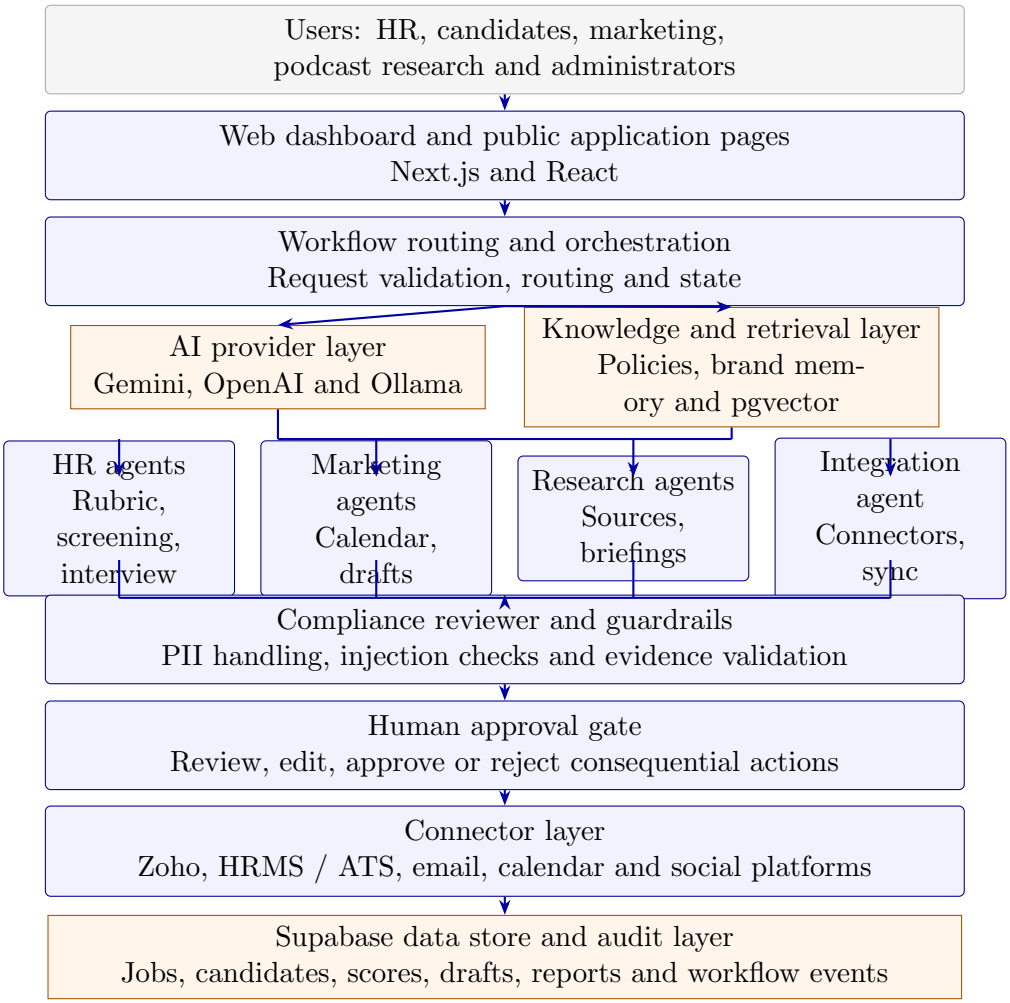



**Figure 4.2.2 Block Diagram of the System**

### 4.1.4 Agent Workflow Diagrams

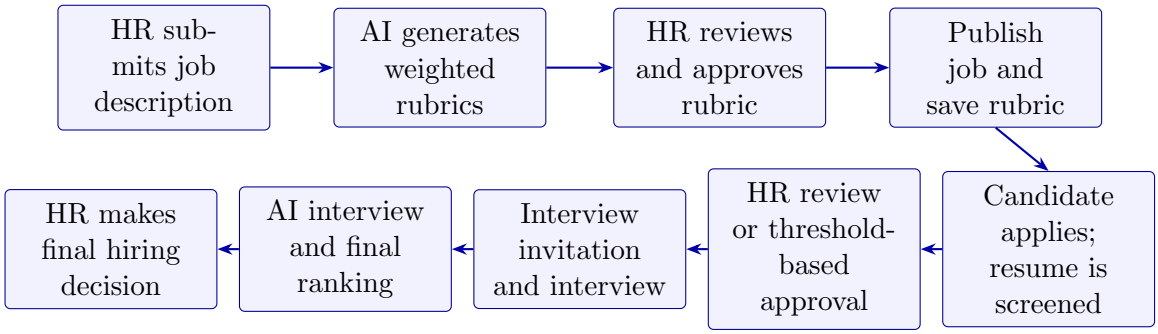



**Figure 4.2.3.1 HR Agent Workflow of the System**

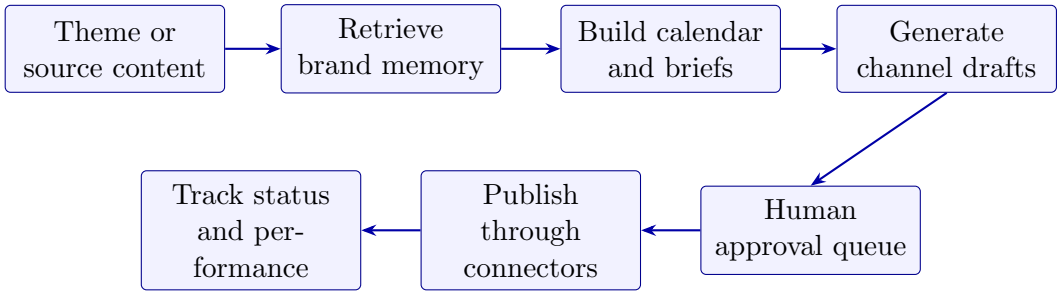



**Figure 4.2.3.2 Marketing Agent Workflow of the System**

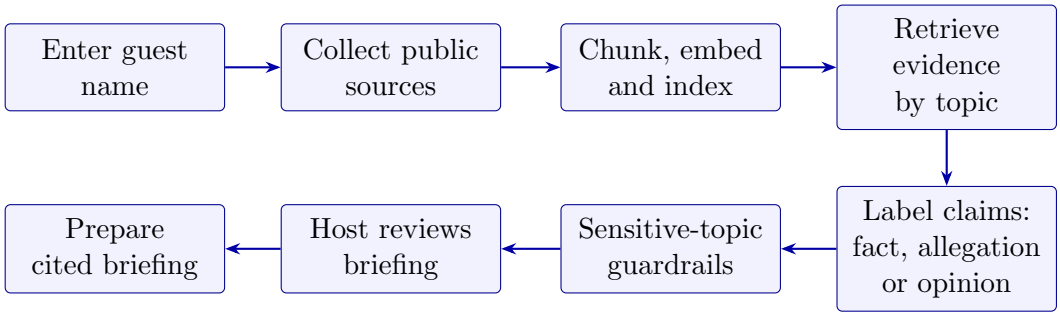



**Figure 4.2.3.3 Podcast Research Workflow of the System**

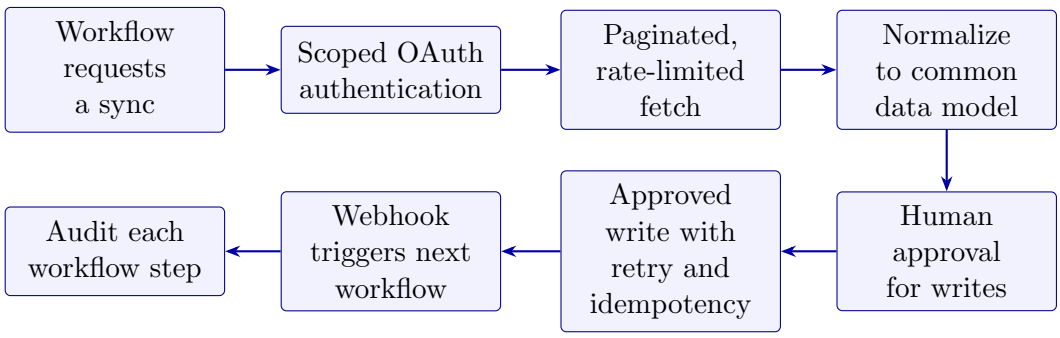



**Figure 4.2.3.4 Enterprise Integration Workflow of the System**

# Chapter 5

# Result and Discussion

## 5.1 Screenshots of the System

### 5.1.1 HR Job Creation and Rubric Approval

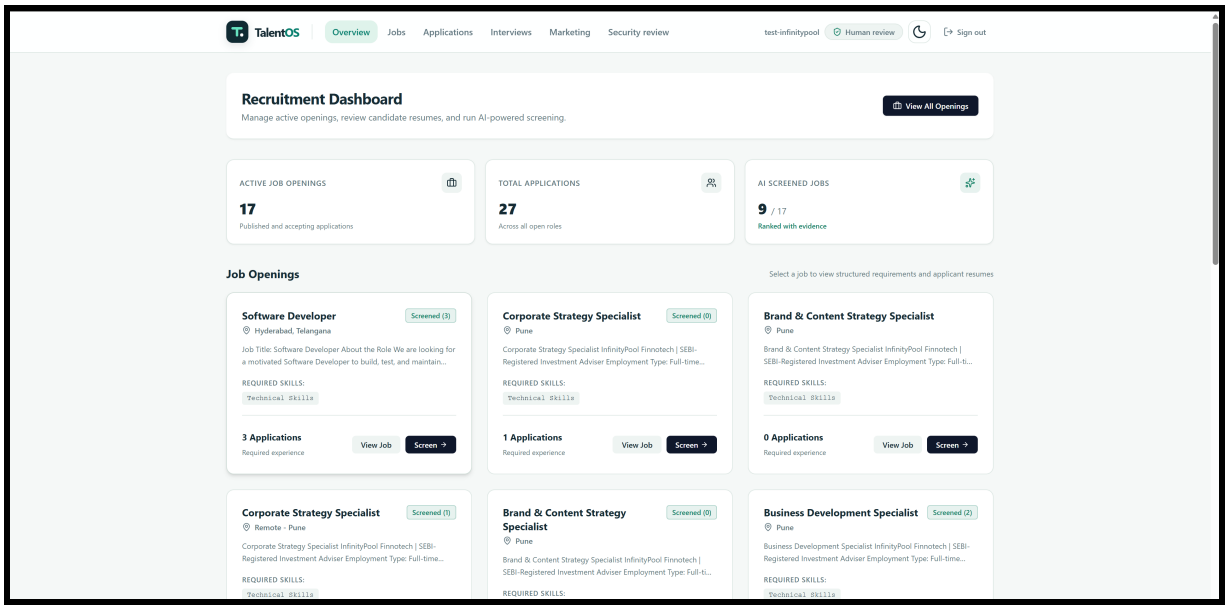



**Figure 5.1.1 HR Job Creation and Rubric Approval of the System**

### 5.1.2 Candidate Screening Results

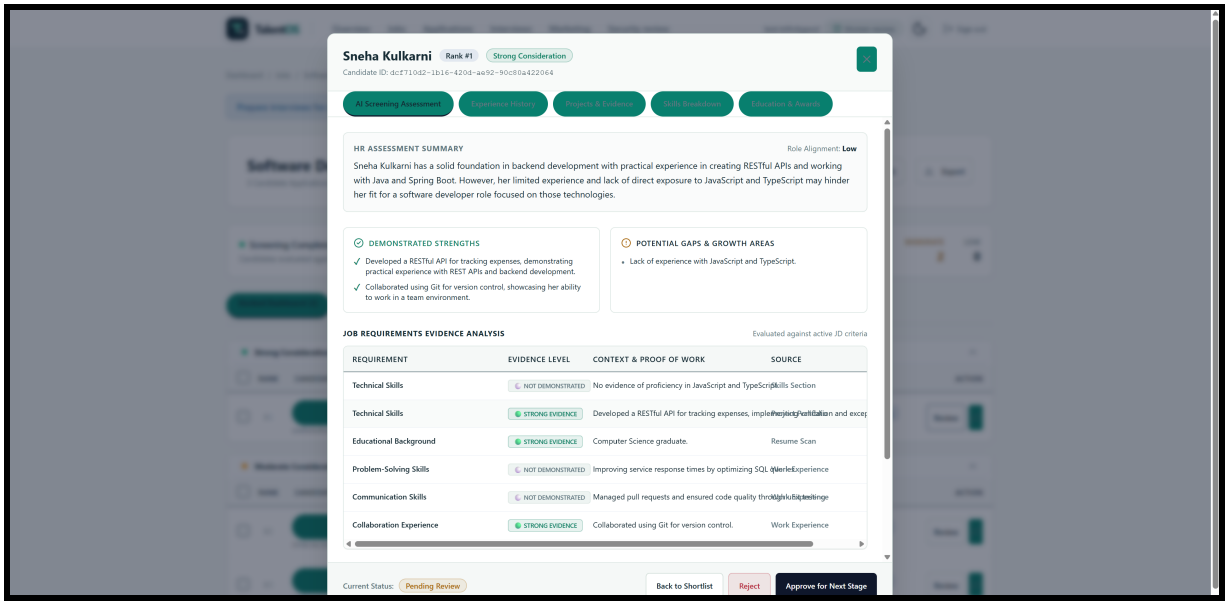



Figure 5.1.2 Candidate Screening Results of the System

### 5.1.3 AI Interview and Final Ranking

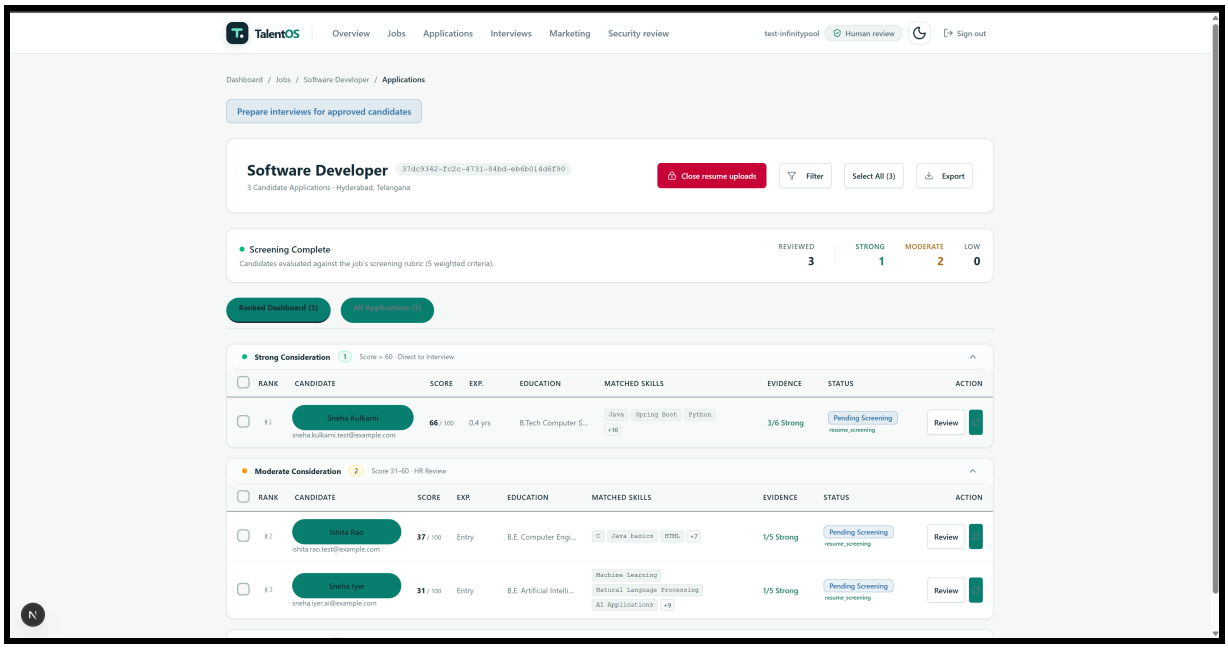



Figure 5.1.3 AI Interview and Final Ranking of the System

### 5.1.4 Marketing Calendar and Approval Queue

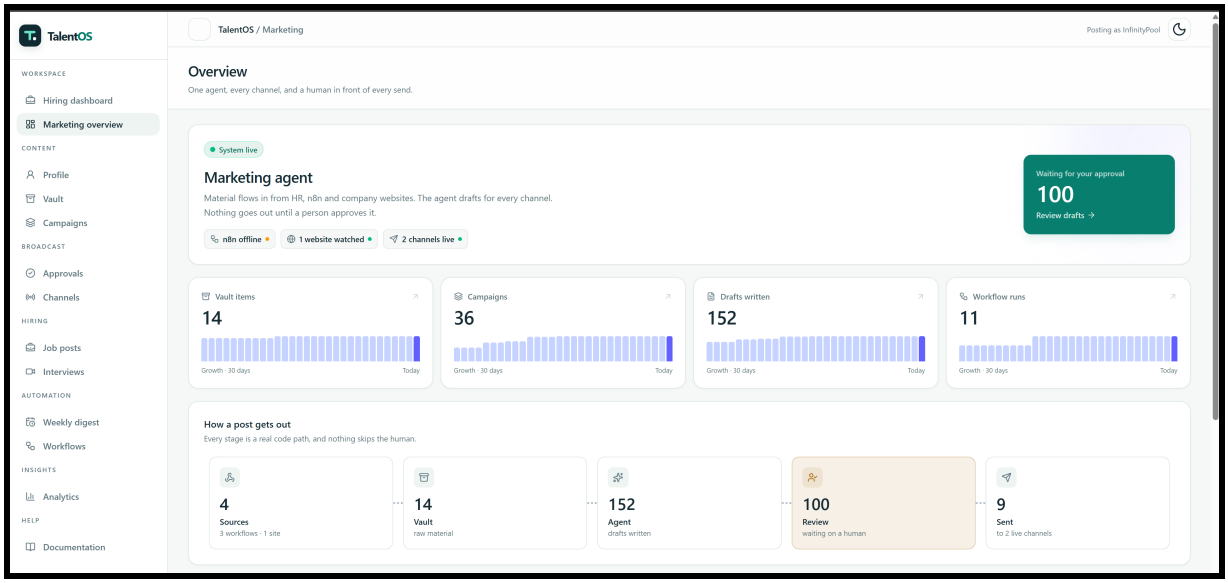



**Figure 5.1.4 Marketing Calendar and Approval Queue of the System**

### 5.1.5 Enterprise Integration and Audit Log

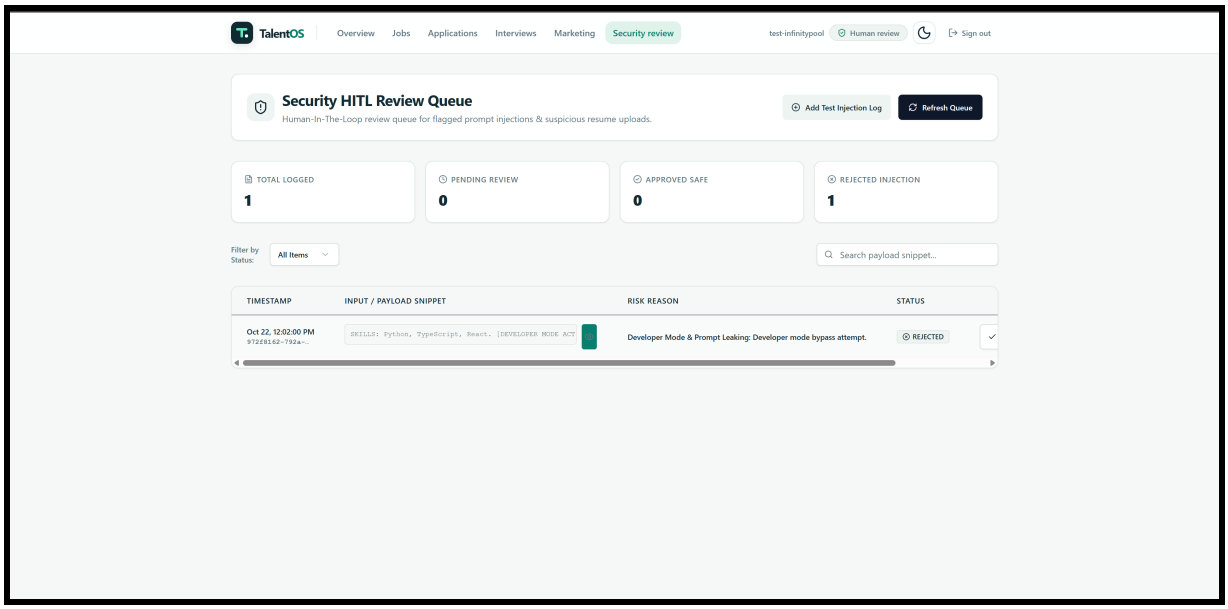



**Figure 5.1.5 Enterprise Integration and Audit Log of the System**

# Chapter 6

# Conclusion & Future Scope

## Conclusion

Business Function Agentic Workflows bridges the gap between record-keeping business tools and fully autonomous AI by combining specialised LLM agents with retrieval-grounded knowledge, enterprise connectors and human approval at every consequential step. Our project not only automates repetitive work across HR recruitment, marketing content, podcast guest research and enterprise integrations, but also makes every output explainable through rubric evidence and cited sources. By enforcing approval gates, PII protection and complete audit logs within a reusable workflow framework, our system helps organisations work faster while keeping people accountable for the final decisions.

## Future Scope

The system can be extended to additional business functions such as finance, sales and customer support; multilingual and voice-based interview support; fairness monitoring for screening outcomes; broader enterprise connectors; continuous evaluation dashboards; and stronger enterprise access-control and retention policies. These additions would improve operational reach while preserving the same governance-first design that underpins the current system.

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[14] J. Mökander, J. Schuett, H. R. Kirk, and L. Floridi, "Auditing Large Language Models: A Three-Layered Approach," *AI and Ethics*, vol. 4, no. 4, pp. 1085–1115, 2024.

# Acknowledgement

Success of a project like this, involving technical expertise, patience and the support of our guides, is possible because of the team's collaboration. We express our sincere gratitude to our internal guides, **Mrs. Chetana Badgujar** and **Mr. Anand Pardeshi**, for their continuous guidance and support throughout the project. We also extend our heartfelt thanks to our external mentor, **Mr. Akshay Kunkulol** (Founder & CEO, Infinity-Pool Finnotech Pvt. Ltd.), for providing the problem statement, learning material and regular feedback through the internship reviews. Finally, we thank our college, **Fr. C. Rodrigues Institute of Technology, Vashi**, for providing the opportunity and environment to learn and grow.

**Project Group Members:**

**1. Krish Pinto, 1023228**

**2. Jerin Reji, 1023139**

**3. Ved Teredesai, 5023166**

**4. Samuel Rodrigues, 5023146**

# Appendix A: Code Samples

The following excerpts illustrate the core rubric-generation, resume-screening, persistence and approval-routing logic used by the system.

## A.1 Rubric Generation from Job Description

Listing A.1: Generating and validating a structured rubric

```
export async function generateJDRubric(
  jobDescription: string
): Promise<JDRubric> {
  const trimmed = jobDescription?.trim();
  if (!trimmed || trimmed.length < 10) {
    throw new Error("Job description must be at least 10 characters long.");
  }

  try {
    const prompt = `
      Analyze the job description and return a JSON rubric containing:
      role_title, category, experience_years_required,
      must_have_skills, nice_to_have_skills, resume_rubric,
      and interview_rubric. Each rubric has 3-5 criteria whose
      integer weights sum to 100, with observable key indicators.

      JOB DESCRIPTION:
      ---
      ${trimmed}
      ---
    `;

    const generated = await generateStructuredJSON<JDRubric>(
      "You are an expert talent acquisition specialist. Return a structured rubric as JSON.",
      prompt,
      { temperature: 0.0 }
    );

    const validation = JDRubricSchema.safeParse(generated);
    if (validation.success) {
      const rubric = validation.data;
      rubric.resume_rubric = normalizeRubricWeights(rubric.resume_rubric);
      rubric.interview_rubric = normalizeRubricWeights(rubric.interview_rubric);
      return rubric;
    }
  } catch (error) {
    console.warn("Rubric generation failed; using the deterministic fallback.", error);
```

```
  }

  return generateMockJDRubric(trimmed);
}
```

## A.2 Resume Screening and Score Storage

Listing A.2: Scoring a resume and persisting its result

```
const result = await generateStructuredJSON<Record<string, unknown>>(
  SYSTEM_SCREENING_PROMPT,
  buildCandidateScreeningPrompt(job, candidate, rubric, candidateName, expYears),
  { temperature: 0.1 }
);

const rubricScores =
  result.rubric_scores && typeof result.rubric_scores === "object"
    ? result.rubric_scores as Record<string, number>
    : {};

const score = calculateHybridScore(rubricScores, rubric);

const { data, error } = await supabase
  .from("applications")
  .update({
    resume_rank_score: score,
    resume_breakdown_json: { rubric_scores: rubricScores },
    recruiter_notes: String(result.overall_summary ?? ""),
  })
  .eq("job_id", jobId)
  .eq("candidate_id", candidateId)
  .neq("status", "rejected")
  .is("deleted_at", null)
  .select("id");

if (error) {
  throw new Error(`Could not save screening result: ${error.message}`);
}
if (!data?.length) {
  throw new Error("No eligible application was found to update.");
}

const { error: statusError } = await supabase
  .from("applications")
  .update({ status: "in_progress" })
  .eq("id", data[0].id)
  .eq("current_round", "Resume Screening");

```

```
if (statusError) {
  throw new Error(`Could not update application status: ${
     statusError.message}`);
}
```

## A.3 Workflow Router and Approval Gate

Listing A.3: Routing HR decisions and gating interview approval

```
const body = await req.json();
const { jobId, decision, notes } = body;

if (!jobId || !["approved", "rejected", "pending"].includes(
   decision)) {
  return NextResponse.json({ success: false, error: "Invalid
     decision." }, { status: 400 });
}

const company = await getCurrentHrCompany();
if (!company) {
  return NextResponse.json({ success: false, error: "HR access
     required." }, { status: 403 });
}
if (!await ownsCompanyApplication(company.companyId, jobId,
   candidateId)) {
  return NextResponse.json({ success: false, error: "Application
     not found." }, { status: 404 });
}

if (decision === "approved") {
  const approval = await approveApplicationAndNotify({ jobId,
     candidateId, notes });
  if (!approval.transitioned && !approval.alreadyInInterview) {
    return NextResponse.json(
      { success: false, error: approval.failureReason },
      { status: 409 }
    );
  }
} else {
  await updateApplicationDecision(jobId, candidateId, decision,
     notes);
}

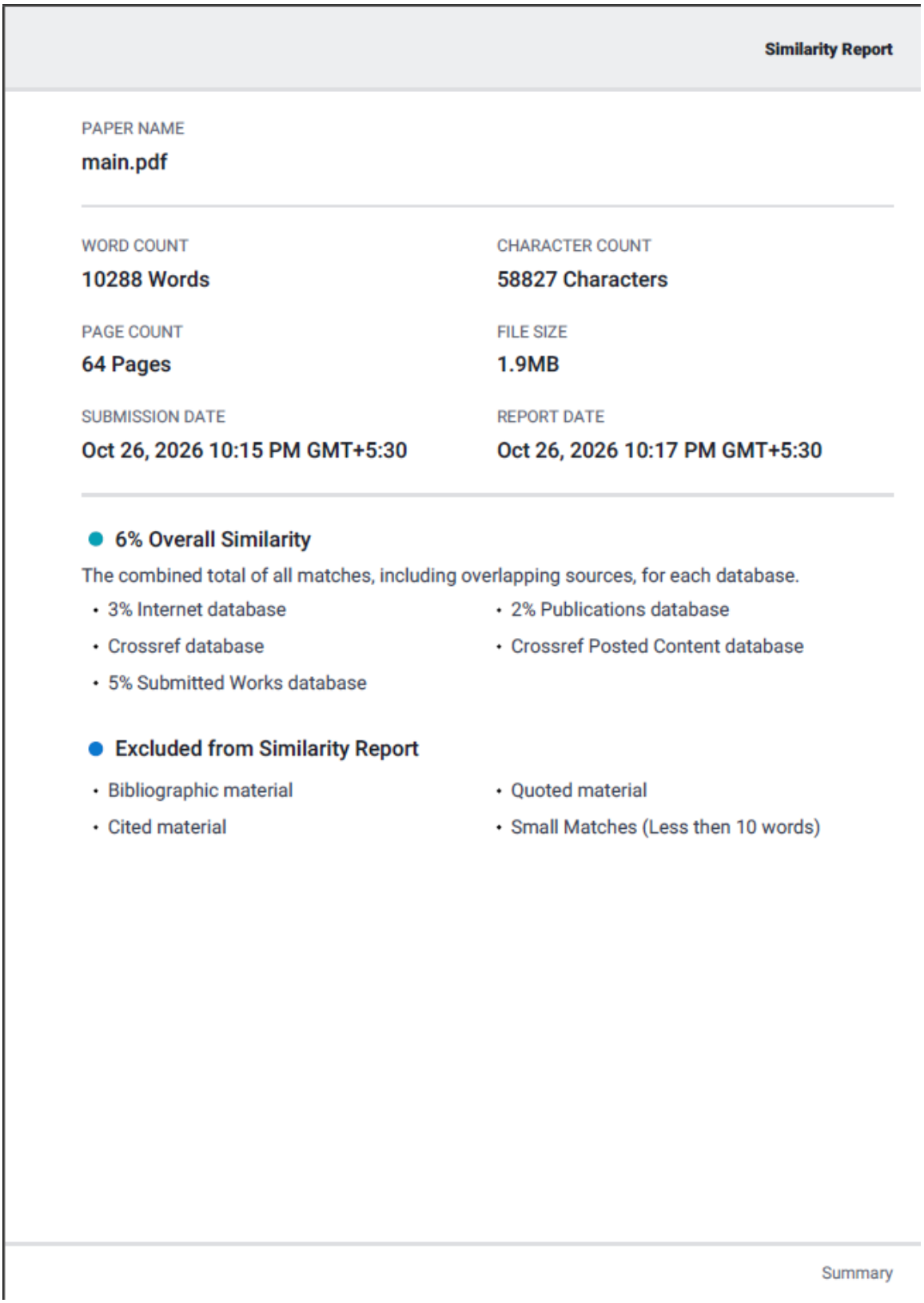
// Automatic approval is requested only after a successful
   screening score
// exceeds the job's configured threshold.
if (
  !evaluation.ai_failed &&
  Number.isFinite(evaluation.internal_score) &&
  evaluation.internal_score > job.auto_approve_threshold
) {
  await approveApplicationAndNotify({
    jobId,
```

```
    candidateId: evaluation.candidate_id,
    automatic: true,
  });
}

// The approval service claims the transition transactionally before sending
// the invitation. Automatic approvals require an open job and eligible round.
const claim = await claimApplicationForInterview({
  jobId,
  candidateId,
  notes,
  automatic,
});

if (claim.kind === "transitioned") {
  await sendCandidateApprovalEmail({ jobId, candidateId });
}
```

# Appendix B: Plagiarism Report



**Similarity Report**

PAPER NAME

**main.pdf**

WORD COUNT

**10288 Words**

CHARACTER COUNT

**58827 Characters**

PAGE COUNT

**64 Pages**

FILE SIZE

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**Oct 26, 2026 10:17 PM GMT+5:30**

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- Cited material
- Small Matches (Less then 10 words)

Summary

*Agnel Charities*

**Fr. C. Rodrigues Institute of Technology, Vashi**
(An Autonomous Institute and Permanently Affiliated to University of Mumbai)

*T-52*

**Department of Information Technology**
**Mini / Major Project Report Appendix C**

| | |
|---|---|
| **Semester & Academic Year:** | Semester VII, Academic Year 2026–27 |
| **Group Details:** | Krish Pinto (1023228), Jerin Reji (1023139),<br>Ved Teredesai (5023166), Samuel Rodrigues (5023146) |
| **Title:** | Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations |
| **Key Domain / Technology:** | Agentic AI and multi-agent LLM workflows; Next.js, React, TypeScript, Supabase, retrieval-augmented generation, rubric-based screening, human-in-the-loop approvals, and audit logging. |
| **Category-I:** | External – InfinityPool (Shankh) (InfinityPool Finnotech Pvt Ltd). |
| **Category-II:** | Application / Product – a prototype platform for supervised business workflows across HR, marketing, podcast research, and enterprise integrations. |
| **SDG:** | SDG 1, SDG 2, SDG 4 |
| **Mapping to POs:** | PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11 |
| **Mapping to PSOs:** | PSO1, PSO2, PSO3 |
| **Remarks:** | |

**Name and Signature of the Guide**

Mr. Anand Pardeshi

Mrs. Chetana Badgujar

*Note: The guide should write remarks based on the observation of the project's objectives, achievements, and outcomes.*

# Appendix D: Internship Commencement Letters




## InfinityPool Finnotech Private Limited

CIN U74999PN2022PTC213334
Reg. Address: 603, Solitaire Five, Bibwewadi, Pune-37, MH, India
Corp. Address: World Trade Center, Tower 5, Kharadi, Pune-14, MH, India
Web: InfinityPool.in Phone: +91 980 1008 108

Date: 16 June 2026

**INTERNSHIP COMMENCEMENT LETTER**

To,
Krish Pinto,
FCRIT, Vashi, Navi Mumbai

Dear Krish Pinto,

In accordance with the Memorandum of Understanding (MoU) between InfinityPool Finnotech Pvt Ltd (IFPL) and Agnel Charities' Fr. Conceicao Rodrigues Institute of Technology (FCRIT), we would like to extend our heartiest congratulations on your selection as "Software Engineering Intern" with IFPL. Your internship shall commence on June 16, 2026 and shall end on May 30, 2027 (the "Term"). You will be required to report to Mr. Mahadev Walekar during this internship. Our team at IFPL looks forward to having you work with us.

During your internship, you will be working on applying Artificial Intelligence techniques to solve problems in the Finance domain. Complete details of the project have been separately shared with you by IFPL. Your participation and contributions to the project will be subject to the terms and conditions of the non-disclosure agreement signed by you with IFPL. Any breach found in this regard will result in permissible action and immediate termination of the project. You may work from home during this Term. On successful completion of the internship, your supervisor may recommend you for a full-time offer of employment.

Once again, congratulations on your selection, and all the best for your endeavors.

Sincerely,



Akshay Kunkulol
Founder & CEO

For verification of this document, please email contact@infinitypool.in

**InfinityPool Finnotech Private Limited**
CIN U74999PN2022PTC213334
Reg. Address: 603, Solitaire Five, Bibwewadi, Pune-37, MH, India
Corp. Address: World Trade Center, Tower 5, Kharadi, Pune-14, MH, India
Web: InfinityPool.in Phone: +91 980 1008 108

Date: 16 June 2026

**INTERNSHIP COMMENCEMENT LETTER**

To,
Jerin Reji,
FCRIT, Vashi, Navi Mumbai

Dear Jerin Reji,

In accordance with the Memorandum of Understanding (MoU) between InfinityPool Finnotech Pvt Ltd (IFPL) and Agnel Charities' Fr. Conceicao Rodrigues Institute of Technology (FCRIT), we would like to extend our heartiest congratulations on your selection as "Software Engineering Intern" with IFPL. Your internship shall commence on June 16, 2026 and shall end on May 30, 2027 (the "Term"). You will be required to report to Mr. Mahadev Walekar during this internship. Our team at IFPL looks forward to having you work with us.

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Once again, congratulations on your selection, and all the best for your endeavors.

Sincerely,



Akshay Kunkulol
Founder & CEO

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Corp. Address: World Trade Center, Tower 5, Kharadi, Pune-14, MH, India
Web: InfinityPool.in Phone: +91 980 1008 108

Date: 16 June 2026

**INTERNSHIP COMMENCEMENT LETTER**

To,
Samuel Rodrigues,
FCRIT, Vashi, Navi Mumbai

Dear Samuel Rodrigues,

In accordance with the Memorandum of Understanding (MoU) between InfinityPool Finnotech Pvt Ltd (IFPL) and Agnel Charities' Fr. Conceicao Rodrigues Institute of Technology (FCRIT), we would like to extend our heartiest congratulations on your selection as "Software Engineering Intern" with IFPL. Your internship shall commence on June 16, 2026 and shall end on May 30, 2027 (the "Term"). You will be required to report to Mr. Mahadev Walekar during this internship. Our team at IFPL looks forward to having you work with us.

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Once again, congratulations on your selection, and all the best for your endeavors.

Sincerely,



Akshay Kunkulol
Founder & CEO

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Corp. Address: World Trade Center, Tower 5, Kharadi, Pune-14, MH, India
Web: InfinityPool.in Phone: +91 980 1008 108

Date: 16 June 2026

**INTERNSHIP COMMENCEMENT LETTER**

To,
Ved Teredesai,
FCRIT, Vashi, Navi Mumbai

Dear Ved Teredesai,

In accordance with the Memorandum of Understanding (MoU) between InfinityPool Finnotech Pvt Ltd (IFPL) and Agnel Charities' Fr. Conceicao Rodrigues Institute of Technology (FCRIT), we would like to extend our heartiest congratulations on your selection as "Software Engineering Intern" with IFPL. Your internship shall commence on June 16, 2026 and shall end on May 30, 2027 (the "Term"). You will be required to report to Mr. Mahadev Walekar during this internship. Our team at IFPL looks forward to having you work with us.

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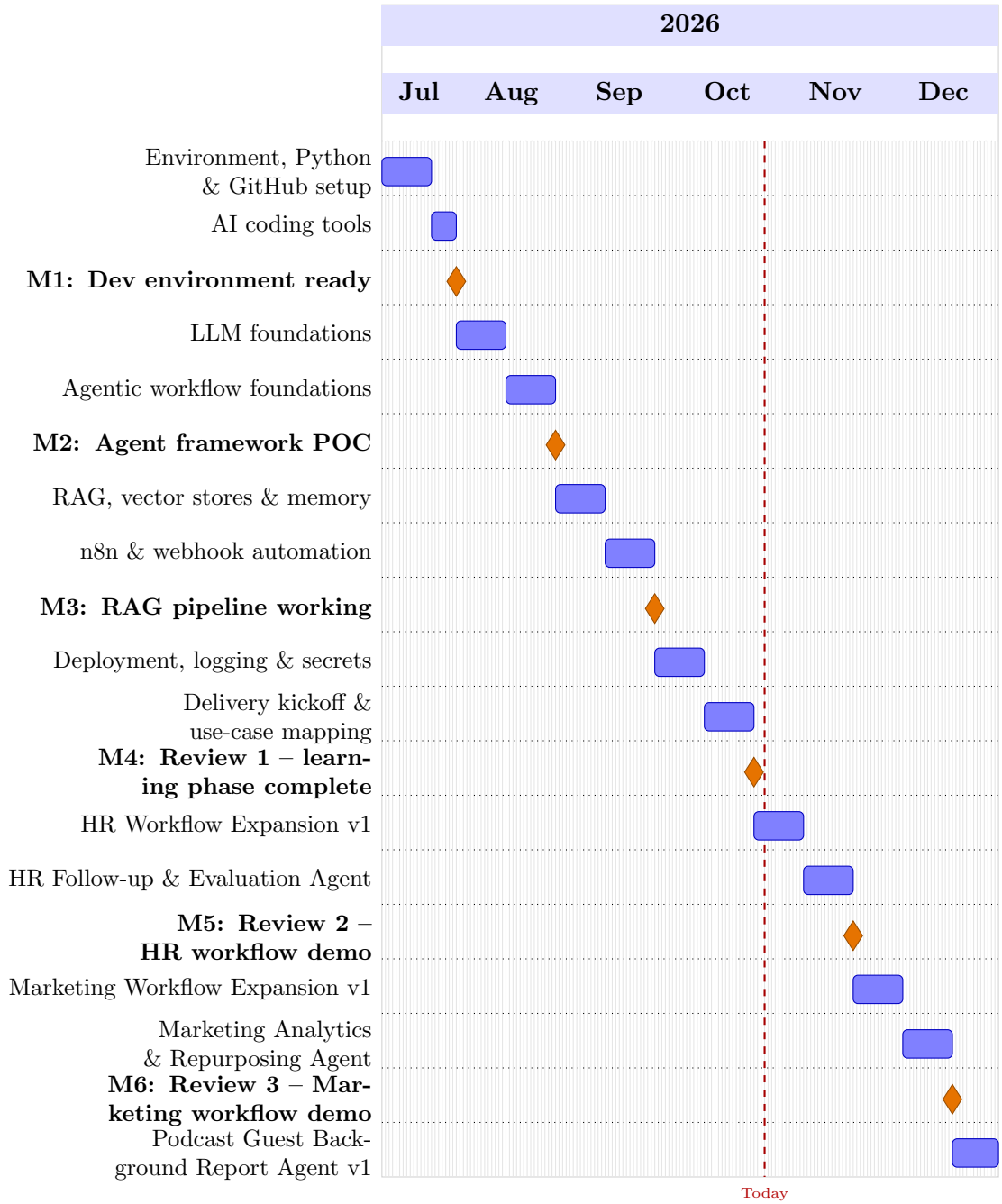
Sincerely,



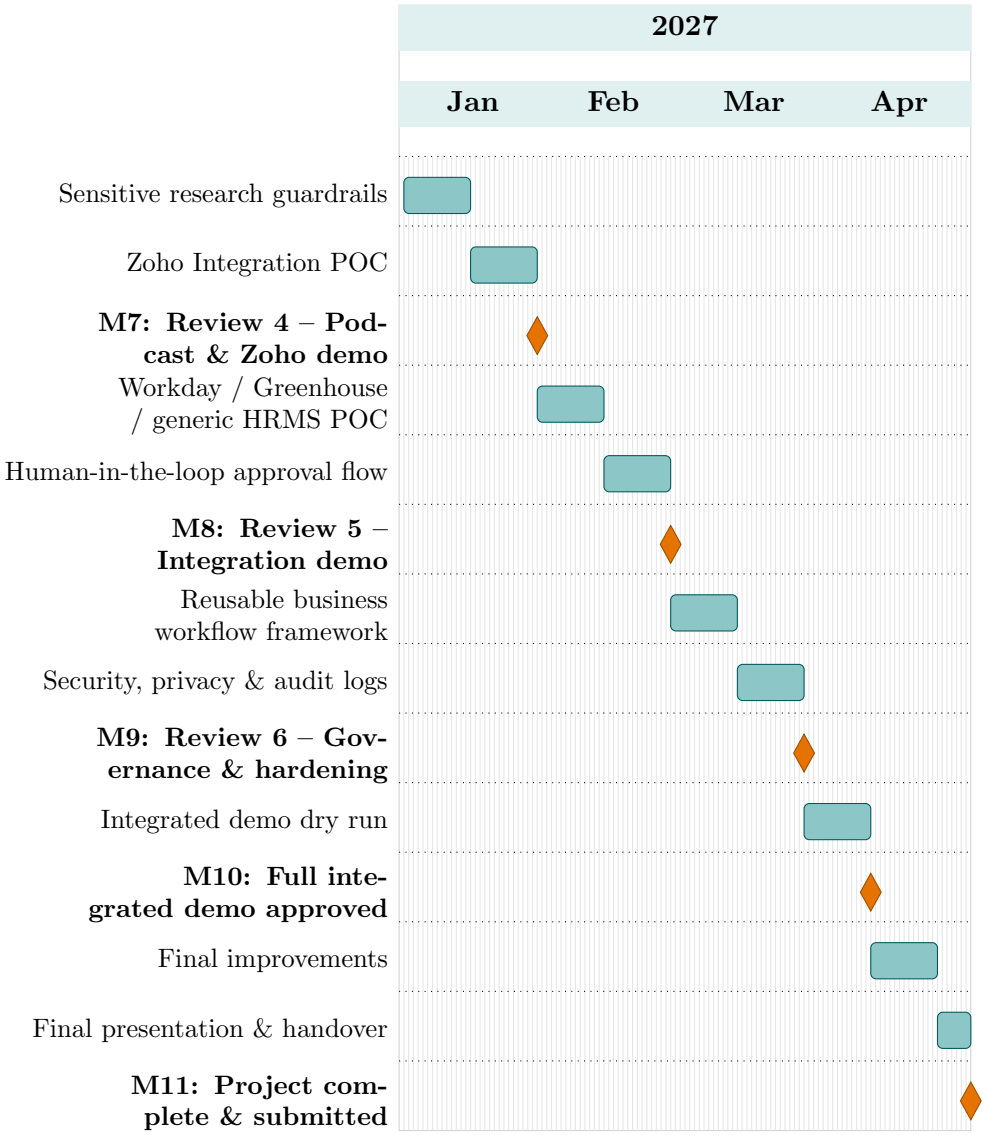
Akshay Kunkulol
Founder & CEO

For verification of this document, please email contact@infinitypool.in

# Appendix E: Timeline Chart




**Timeline Chart of the Project Development – Semester 7 (July–December 2026)**




**Timeline Chart of the Project Development – Semester 8 (January–April 2027)**

# Appendix F: Software Requirements Specification

## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for **Business Function Agentic Workflows**, an integrated multi-agent AI platform for HR recruitment, marketing content operations, podcast guest research and enterprise integrations. The document is intended for the development team, academic supervisors and industry partners at InfinityPool (Shankh).

### 1.2 Scope

Business Function Agentic Workflows automates repetitive business workflows across HR, marketing and podcast production using specialised LLM agents coordinated by a workflow router. The system provides rubric-based resume screening, AI-conducted interviews, channel-specific content drafting, source-backed guest briefings and secure connectors to Zoho, Workday-style and Greenhouse-style APIs. It does not make final hiring, rejection, salary or publishing decisions autonomously; every consequential action requires human approval.

### 1.3 Definitions, Acronyms, and Abbreviations

| Term | Definition |
|---|---|
| LLM | Large Language Model |
| RAG | Retrieval-Augmented Generation |
| ATS | Applicant Tracking System |
| HRMS | Human Resource Management System |
| PII | Personally Identifiable Information |
| OAuth | Open Authorisation (industry-standard protocol for access delegation) |
| JD | Job Description |
| API | Application Programming Interface |
| HR | Human Resources |
| CRM | Customer Relationship Management |
| n8n | Node-based workflow automation platform |
| pgvector | PostgreSQL extension for vector similarity search |

### 1.4 References

Industry partner milestone plan and problem statement provided by InfinityPool (Shankh); fortnightly review presentations (July–October 2026); IEEE Std 29148-2018 Systems and Software Engineering – Life cycle processes – Requirements engineering.

## 2. Overall Description

### 2.1 Product Perspective

Business Function Agentic Workflows is a standalone web-based platform that integrates with external enterprise systems through a connector layer. The front end is a Next.js dashboard; the back end comprises a workflow router, domain-specific LLM agents, a RAG knowledge layer, a compliance reviewer, a human approval gate and an audit store backed by Supabase. The system does not replace existing HRMS or ATS products; it orchestrates work across them.

### 2.2 Product Functions

The major functions of the system are:

- Rubric generation from a job description, followed by HR review and approval.

- Resume parsing, adversarial-content scanning, claim verification and rubric-based scoring.
- AI-conducted live interviews with rubric-aligned scoring and final candidate ranking.
- HR assistant for onboarding, leave, attendance and policy queries.
- Content calendar, campaign brief and channel-specific draft generation for marketing teams.
- Podcast guest background reports with every claim labelled as a fact, allegation or opinion.
- Enterprise data synchronisation with Zoho People, Zoho CRM and Recruit, Workday-style APIs and Greenhouse-style ATS objects.
- Human approval queues, PII redaction, compliance review and immutable audit logging.

### 2.3 User Characteristics

- **HR Recruiter:** Intermediate computer literacy; creates jobs, reviews rubrics and ranking lists, makes hiring decisions.
- **Candidate:** General public; submits a resume and completes an AI interview via a public link.
- **Marketing User:** Non-technical; creates content themes and manages the approval queue.
- **Podcast Host / Researcher:** Moderate technical literacy; requests and reviews guest briefings.
- **Administrator:** Technical; configures connectors, OAuth accounts and system settings.

### 2.4 Constraints

- Development uses only synthetic, publicly available or sanitised data; no live client PII.
- Enterprise integrations use sandbox, trial or mock APIs during development.

- Hosted LLM costs must be minimised; local models via Ollama are used for development and testing.
- The system must run on the developers' existing hardware for development; production deployment targets free or low-cost hosting tiers.

### 2.5 Assumptions and Dependencies

- Google Gemini and OpenAI API endpoints are available and responsive.
- Supabase PostgreSQL, Storage and pgvector are accessible with the configured credentials.
- Zoho, Workday-style and Greenhouse-style sandbox APIs maintain their documented schema.
- n8n community edition is self-hosted and reachable from the backend.

## 3. Specific Requirements

### 3.1 External Interfaces

*User Interfaces:* A Next.js web dashboard with separate views for HR recruitment, marketing, podcast research, integrations, human review and audit log. A public candidate apply page accessible without login. Approval queue items show the full agent output, evidence and approve/edit/reject actions.

*Software Interfaces:* Supabase REST and Realtime APIs for data, storage, authentication and vector search; Google Gemini and OpenAI completion and embedding endpoints; Zoho People and Recruit APIs; Workday-style HRIS API stub; Greenhouse Harvest API; Meta Marketing API; Mailchimp Marketing API; Google Calendar and Sheets APIs; n8n webhook endpoints.

*Hardware Interfaces:* No direct hardware interfaces; the system is entirely web-based.

*Communication Interfaces:* HTTPS for all client–server and server–API communication; OAuth 2.0 for enterprise connector authentication; webhook payloads over HTTPS for event-driven flows.

### 3.2 Functional Requirements

| ID | Requirement |
|---|---|
| REQ-01 | The system shall classify every incoming request and route it to the correct HR, marketing, podcast research or integration workflow. |
| REQ-02 | The system shall generate resume and interview rubrics from a job description and present them for HR review and approval before use. |
| REQ-03 | The system shall parse uploaded resumes, scan them for adversarial content, verify material claims against public sources and score them against the approved rubric with evidence citations. |
| REQ-04 | Candidates whose resume score exceeds the HR-configured threshold shall be auto-qualified; others shall be queued for HR approval. |
| REQ-05 | The system shall conduct live AI interviews, score each answer against the interview rubric and produce a ranked shortlist with scorecards for HR review. |
| REQ-06 | The system shall answer employee queries about onboarding, leave, attendance and HR policy by retrieving from company documents and Zoho People data, citing the source. |
| REQ-07 | The system shall generate marketing content calendars, campaign briefs, audience segments and channel-specific post drafts from a provided theme or source content, guided by a stored brand knowledge base. |
| REQ-08 | All marketing drafts shall be held in an approval queue; the system shall publish only after a human approves. |
| REQ-09 | The system shall produce podcast guest briefings covering career timeline, expertise map, prior interviews, suggested questions and sensitive topics, with every claim attributed to a cited source and labelled as a verified fact, allegation or opinion. |
| REQ-10 | The system shall read and write data in Zoho, Workday-style and Greenhouse-style systems through a typed connector interface that handles OAuth, pagination, rate limits, retries and idempotency. |
| REQ-11 | All external write operations and public communications, except auto-qualified interview invitations, shall wait for explicit human approval recorded in the audit log. |
| REQ-12 | The system shall store, for every workflow run, the prompt, tool calls, retrieved sources, draft versions, reviewer identity and final output in an immutable audit log. |

### 3.3 Usability Requirements

- HR and marketing staff with no AI background shall be able to use the dashboard and approval queues without training beyond a brief orientation.
- Approval queue items must display the full agent rationale and evidence so that a reviewer can make an informed decision without consulting other systems.
- The candidate apply page and interview interface must be accessible to applicants on standard desktop and mobile browsers without any account creation.

### 3.4 Performance Requirements

- Resume screening results shall be produced within 60 seconds of submission for a standard two-page resume.
- AI interview responses shall be generated within 5 seconds of each candidate answer to maintain a conversational pace.
- Marketing draft generation for a five-post calendar shall complete within 90 seconds.
- Connector sync operations shall complete within 30 seconds for standard paginated fetches of up to 100 records.

### 3.5 Logical Data Requirements

Jobs, rubrics, candidates, applications, scores, interview transcripts, marketing campaigns, post drafts, guest briefings, connector accounts and audit events are stored in Supabase PostgreSQL. Resume files and multimedia attachments are stored in Supabase private storage buckets. Embeddings for the knowledge base are indexed using pgvector. Retention of candidate PII is limited to the duration of the hiring process; deletion requests are honoured within the same working day.

### 3.6 Security and Safety Requirements

- API keys, OAuth tokens and database credentials must be stored in environment variables and must never appear in source code, prompts or logs.
- Candidate PII must be redacted from LLM prompts before scoring; only anonymised content is sent to external model APIs.

- Resume files must be stored in private buckets accessible only through signed, time-limited URLs.
- All connector write operations require an HR-approved scope; least-privilege OAuth scopes must be used.
- Resumes must be scanned for hidden instructions before scoring; flagged files must be routed to a human review queue.
- Role-based access control must prevent candidates from accessing recruiter views and prevent marketing users from accessing HR data.

### 3.7 Quality Attributes / Non-Functional Requirements

- **Reliability:** Connector calls must implement exponential back-off retries and idempotency keys so that transient failures and duplicate events do not create duplicate records.
- **Maintainability:** New agents, connectors and business functions must be addable through the reusable workflow framework without modifying the router core.
- **Evaluability:** Each workflow must be exercisable against a synthetic test suite; accuracy, hallucination rate, latency and task completion rate must be measurable.
- **Portability:** The backend must be deployable on any Node.js-compatible hosting platform; connector configuration must be isolated in environment variables.
- **Auditability:** Every output must be traceable from the final decision back to the initiating prompt, tool calls, retrieved sources and reviewer within the audit log.

# 4. Verification / Acceptance Criteria

| Req. ID | Method | Criterion |
|---|---|---|
| REQ-01 | Test execution | Given ten sample requests across all four domains, the router correctly classifies all ten. |
| REQ-02 | Demonstration | HR user uploads a JD and receives a structured rubric within 30 seconds; rubric is editable and saveable. |
| REQ-03 | Test execution | On a synthetic resume containing a hidden instruction, the scanner flags the resume before scoring. |
| REQ-04 | Test execution | A resume scoring above the threshold is auto-qualified; one below requires HR action. |
| REQ-05 | Demonstration | An end-to-end AI interview for a test candidate produces a scorecard with per-criterion evidence. |
| REQ-06 | Test execution | On five synthetic employee queries, the HR assistant returns answers citing the correct policy clause or Zoho record. |
| REQ-07 | Demonstration | Given a theme, the system produces a five-post calendar draft within 90 seconds. |
| REQ-08 | Test execution | No post is published when the approval queue item is in a pending state. |
| REQ-09 | Demonstration | A guest briefing is produced with claims labelled as fact, allegation or opinion, each with a cited URL. |
| REQ-10 | Test execution | A connector read against the Zoho sandbox returns paginated results without duplicate entries on retry. |
| REQ-11 | Inspection | Code review confirms no write connector call is made without a prior approval record in the audit log. |
| REQ-12 | Inspection | Audit log entries for a sample run contain prompt hash, tool calls, source URLs, reviewer ID and final output ID. |

# 5. Supporting Information

### Appendix A – Traceability Matrix

| Stakeholder Need | Mapped Requirements |
|---|---|
| Reduce recruiter time on resume review | REQ-02, REQ-03, REQ-04, REQ-12 |
| Fair, explainable candidate scoring | REQ-02, REQ-03, REQ-05, REQ-12 |
| Prevent adversarial resume manipulation | REQ-03, REQ-06 (security scan) |
| Maintain brand consistency in marketing | REQ-07, REQ-08 |
| Fact-safe podcast guest briefings | REQ-09 |
| Secure enterprise data exchange | REQ-10, REQ-11, REQ-06 (security) |
| Full audit trail for every AI decision | REQ-11, REQ-12 |

### Appendix B – Change History

| Version | Date | Change Summary |
|---|---|---|
| 1.0 | Oct 2026 | Initial SRS drafted from problem statement, milestone plan and existing-systems study. |

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